Python Activity-Based Learning

Case Study 4: Employee Payroll and Deduction Analysis

1. Problem Statement

Calculate an employee's gross salary from basic pay, allowances, overtime, and incentives. Apply progressive deductions and print a formatted salary statement.

2. Objectives

- Use variables and arithmetic operators to calculate salary components.
- Apply progressive deductions with if/elif/else conditions.
- Display a readable salary statement.

3. Assumptions

- HRA is 20% of basic pay and travel allowance is 5% of basic pay.
- Overtime is paid at ₹150 per hour; incentive is entered by the user.
- Deduction is 5% of gross salary up to ₹25,000; 10% on the portion above ₹25,000 up to ₹50,000; and 15% on the portion above ₹50,000.

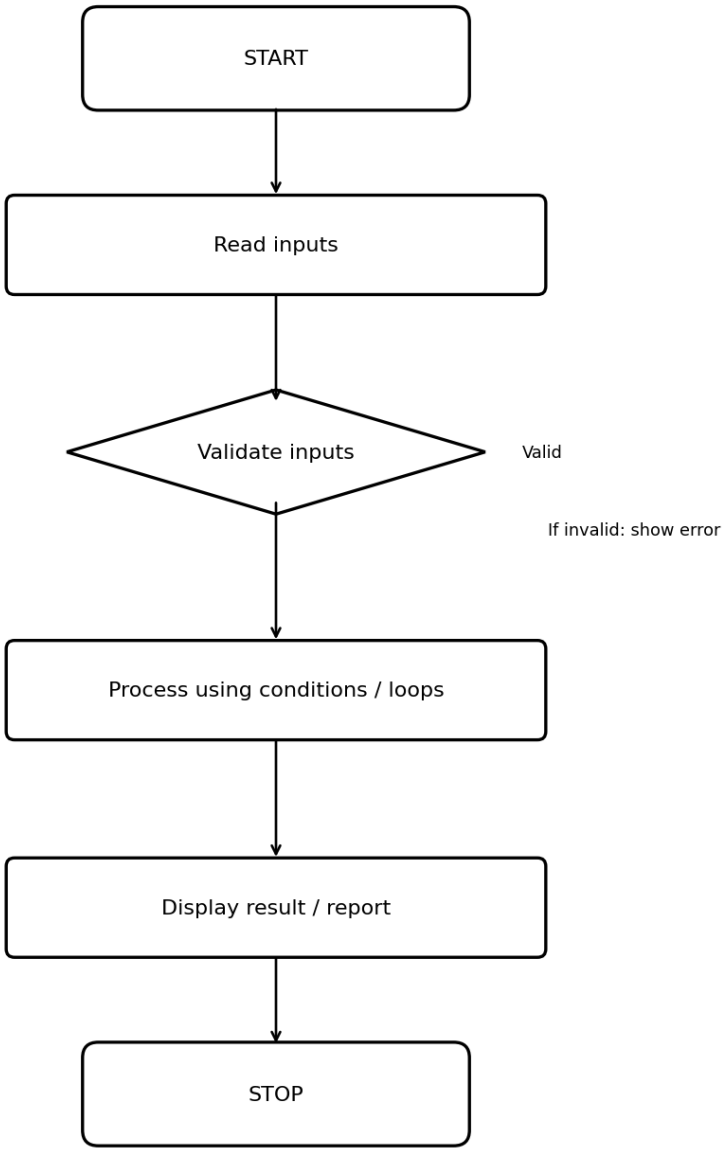
4. Algorithm

1. Start and read employee name, basic pay, overtime hours, and incentive.
2. Validate that numeric inputs are non-negative.
3. Calculate HRA, travel allowance, overtime pay, and gross salary.
4. Calculate progressive deduction by salary bands.
5. Calculate net salary as gross salary minus deduction.
6. Print the salary statement and stop.

5. Flowchart

The diagram below shows the overall program flow. The validation decision represents the input checks described in the algorithm.

Flowchart: Employee Payroll and Deduction Analysis



6. Pseudocode

```
START
READ name, basic_pay, overtime_hours, incentive
IF any numeric input is negative THEN DISPLAY error
ELSE
    hra = 20% of basic_pay
    travel = 5% of basic_pay
    overtime_pay = overtime_hours * 150
    gross = basic_pay + hra + travel + overtime_pay + incentive
    IF gross <= 25000 THEN deduction = 5% of gross
    ELSE IF gross <= 50000 THEN deduction = 1250 + 10% of (gross - 25000)
    ELSE deduction = 3750 + 15% of (gross - 50000)
    net = gross - deduction
    DISPLAY salary statement
```

```
END IF
STOP
```

7. Python Source Code

The solution uses only basic Python features from the assigned syllabus.

```
name = input("Employee name: ")
basic = float(input("Basic pay (₹): "))
overtime_hours = float(input("Overtime hours: "))
incentive = float(input("Incentive (₹): "))

if basic < 0 or overtime_hours < 0 or incentive < 0:
    print("Invalid input: values cannot be negative.")
else:
    hra = basic * 0.20
    travel = basic * 0.05
    overtime_pay = overtime_hours * 150
    gross = basic + hra + travel + overtime_pay + incentive

    if gross <= 25000:
        deduction = gross * 0.05
    elif gross <= 50000:
        deduction = 1250 + (gross - 25000) * 0.10
    else:
        deduction = 3750 + (gross - 50000) * 0.15

    net = gross - deduction
    print("\n------ SALARY STATEMENT ------")
    print("Employee:", name)
    print(f"Basic pay       : ₹{basic:.2f}")
    print(f"HRA             : ₹{hra:.2f}")
    print(f"Travel allowance: ₹{travel:.2f}")
    print(f"Overtime pay    : ₹{overtime_pay:.2f}")
    print(f"Incentive       : ₹{incentive:.2f}")
    print(f"Gross salary    : ₹{gross:.2f}")
    print(f"Deduction       : ₹{deduction:.2f}")
    print(f"Net salary      : ₹{net:.2f}")
```

8. Test Cases

Test input / condition	Expected result	Purpose
20000, 2, 1000	Basic ₹20,000; 2 OT hours; ₹1,000 incentive	Gross ₹26,000; deduction ₹1,350; net ₹24,650
40000, 0, 0	Basic ₹40,000; no overtime/incentive	Gross ₹50,000; deduction ₹3,750; net ₹46,250
-1, 0, 0	Negative basic pay	Invalid input message

9. Sample Output Screenshot

Sample console output for one test run:

```
C:\Users\Badri\PycharmProjects\PythonProject-1\.venv\Scripts\python.exe "C:\Users\Badri\PycharmProjects\PythonProject-1\Case Studies\case
 study -5.py"
Employee name: Kiran
Basic pay (₹): 20000
Overtime hours: 2
Incentive (₹): 1000

------ SALARY STATEMENT ------
Employee: Kiran
Basic pay       : ₹20000.00
HRA             : ₹4000.00
Travel allowance: ₹1000.00
Overtime pay    : ₹300.00
Incentive       : ₹1000.00
Gross salary    : ₹26300.00
Deduction       : ₹1380.00
Net salary      : ₹24920.00

Process finished with exit code 0
```

10. Conclusion

The solution demonstrates salary calculations, progressive deductions, validation, and formatted output using basic Python constructs.

Note: Assumed rates, thresholds, and sample values are for this educational exercise and can be changed if the instructor supplies different rules.